How INLA works ANSWER KEY

Three easy steps for one county

Work in pairs. Keep a phone calculator handy.

The problem

Pine County had **9 deaths** where **5** were expected: SMR = $9/5 = 1.8$. That rests on very few deaths, so, like Restaurant A, Pine County should **borrow** from the rest of the state. **But how much?** Nobody knows. So INLA tries a few amounts and takes an average.

Step 1. Try three amounts of borrowing

4 minutes

Borrowing k means: add k deaths to the observed *and* k to the expected, then divide.

$$\text{best guess} = \frac{9+k}{5+k}$$

	Borrow k	Best guess
A borrow a little	5	$\frac{9+5}{5+5} = \frac{14}{10} = 1.40$
B borrow some	20	**29/25 = 1.16**
C borrow a lot	100	**109/105 = 1.04**

1. The more we borrow, the closer the best guess gets to **1**, the state average.

Step 2. Average them

4 minutes

INLA does not pick one guess. It averages all three, giving more weight to the amount of borrowing that fits the data of **all the counties** best. R finds these weights for us:

	A a little	**B** some	**C** a lot
Weight	0.25	0.60	0.15

It works just like your **course grade**: homework 25%, exam 60%, quizzes 15%.

2. Pine County's final best guess:

$$0.25 \times 1.40 + 0.60 \times \textbf{1.16} + 0.15 \times \textbf{1.04} = \textbf{1.20}$$

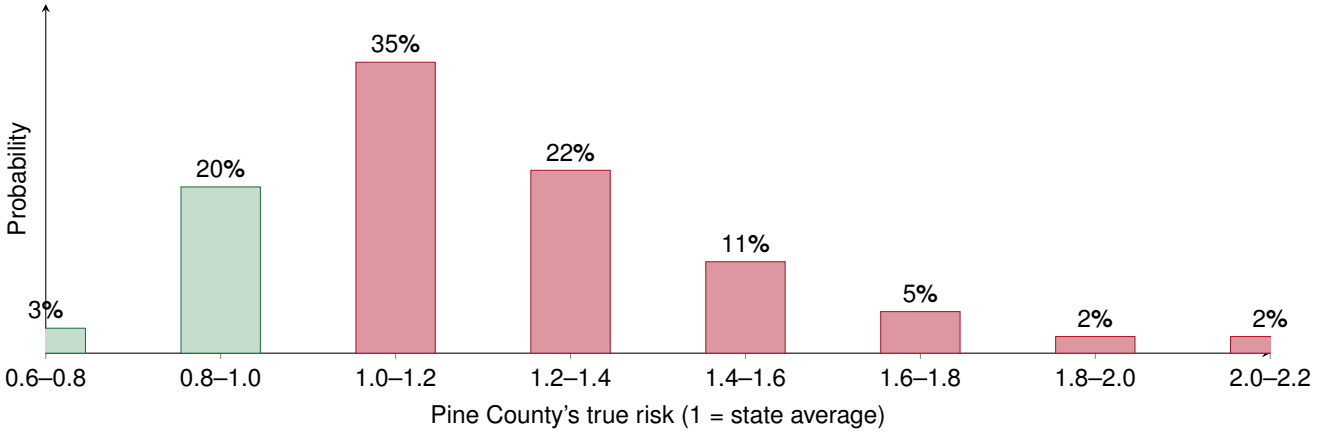
3. Why average the three instead of keeping only B?

- ■ Because we are not sure how much to borrow.
- ☐ Because B is wrong.
- ☐ Because averaging always gives 1.

Step 3. Read the answer

6 minutes

INLA's final answer is not one number but a whole range of possible values, each with a probability. For Pine County:



4. How likely is Pine County above the state average? Add up the bars to the right of 1.0:

__35__ + __22__ + __11__ + __5__ + __2__ + __2__ = __77%__

This is the **exceedance probability**.

5. The 95% credible interval. Throw away the lowest 2.5% and the highest 2.5%. The lowest bar (3%) and the two highest bars (2% + 2%) hold just about that much, so the cut points are near the edges of those bars. Pine County's risk is between about **0.8** and __1.9__, with 95% probability. *(1.9 or 2.0 are both fine.)*

6. Pine County's SMR was 1.8. Is it clearly a high-risk county? Use questions 4 and 5. *Not clearly. The best guess is only 1.20, the interval (0.8 to 1.9) includes 1, and there is a 23% chance it is not above average. Nine deaths are not enough to be sure.*

In R

INLA does Steps 1 to 3 for every county at once. You read the results here:

On this handout	In R, after `m <- inla(...)`
best guess (Step 2)	`m$summary.fitted.values$mean`
95% credible interval (question 5)	`0.025quant` and `0.975quant` columns
probability above 1 (question 4)	`1 - inla.pmarginal(1, marg)`